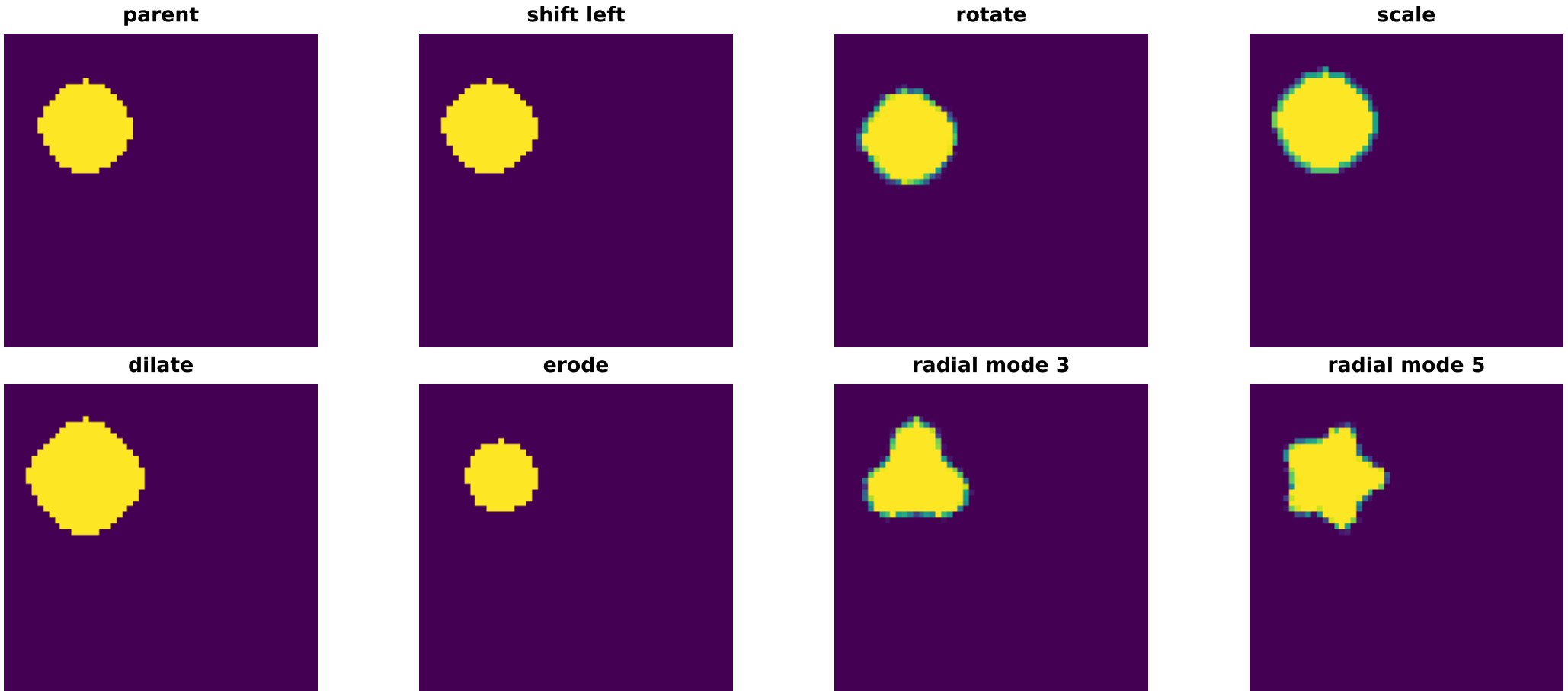A single packet generates an unbounded perturbation family